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In [5]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from itertools import product

print("="*70)
print(" FACTORIAL ANALYSIS: MAIN & INTERACTION EFFECTS (PARETO CHART)")
print("="*70)

# 1. Generate the 2^4 Full Factorial Design Matrix (-1, 1)
factors = ['Temp (T)', 'Moist (M)', 'Size (PS)', 'Rate (R)']
design = list(product([-1, 1], repeat=4))
df = pd.DataFrame(design, columns=factors)

# 2. Generate 2-Way Interaction Columns
interactions = []
for i in range(len(factors)):
    for j in range(i + 1, len(factors)):
        col_name = f"{factors[i].split(' ')[1].strip('(')}x{factors[j].split(' ')[1].strip('(')}"
        df[col_name] = df[factors[i]] * df[factors[j]]
        interactions.append(col_name)

all_terms = factors + interactions

# 3. Simulate Plant PDI Results
# (We inject a massive negative effect for Rate, a moderate one for Size/Temp,
# and a strong T*M interaction to simulate steam conditioning physics)
np.random.seed(42)
df['Achieved_PDI'] = (
    90.0
    - 5.25 * df['Rate (R)']
    - 1.60 * df['Size (PS)']
    - 1.49 * df['Temp (T)']
    + 0.48 * df['Moist (M)']
    - 2.10 * df['TxM'] # The crucial steam conditioning interaction
    + np.random.normal(0, 0.3, 16) # Factory noise
)

# --- NEW: DISPLAY THE SIMULATED RAW DATA ---
print("\n" + "-"*65)
print(" SIMULATED 16-RUN FACTORIAL MATRIX (RAW FACTORY DATA)")
print("-"*65)
# Filter out the interaction columns just to keep the terminal printout clean
display_df = df[['Temp (T)', 'Moist (M)', 'Size (PS)', 'Rate (R)', 'Achieved_PDI']]
display_df['Achieved_PDI'] = display_df['Achieved_PDI'].round(2) # Round to 2 decimal places
print(display_df.to_string(index=False))
print("-"*65 + "\n")

# 4. Calculate the Factorial Effects
# Formula: Effect = Mean(PDI at +1) - Mean(PDI at -1)
effects = {}
for term in all_terms:
    mean_high = df[df[term] == 1]['Achieved_PDI'].mean()
    mean_low = df[df[term] == -1]['Achieved_PDI'].mean()
    effects[term] = mean_high - mean_low

# 5. Format for Pareto Chart
df_effects = pd.DataFrame(list(effects.items()), columns=['Term', 'Effect'])
df_effects['Abs_Effect'] = df_effects['Effect'].abs()
df_effects = df_effects.sort_values(by='Abs_Effect', ascending=True) # Sort for horizontal bar chart

print("Calculated Factorial Effects (Ranked by Magnitude):")

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print("-" * 50)
for index, row in df_effects.sort_values(by='Abs_Effect', ascending=False).iterrows():
    direction = "Decreases" if row['Effect'] < 0 else "Increases"
    print(f"{row['Term']:<10} : {row['Abs_Effect']:>5.2f}% PDI swing ({direction})")

# 6. Plot the Pareto Chart
plt.figure(figsize=(10, 6))
colors = ['#e74c3c' if val < 0 else '#2ecc71' for val in df_effects['Effect']]

bars = plt.barh(df_effects['Term'], df_effects['Abs_Effect'], color=colors, edgecolor='black')

plt.axvline(x=0.5, color='black', linestyle='--', alpha=0.5, label='Noise Threshold')

plt.title('Pareto Chart of Standardized Effects (2^4 Factorial Design)', fontsize=12)
plt.xlabel('Absolute Effect on Pellet Durability Index (%)', fontweight='bold')
plt.ylabel('Process Variables & Interactions', fontweight='bold')

# Add value labels to bars
for bar, effect in zip(bars, df_effects['Effect']):
    plt.text(bar.get_width() + 0.1, bar.get_y() + bar.get_height()/2,
             f"{effect:+.2f}", va='center', fontweight='bold')

# Custom Legend
from matplotlib.patches import Patch
legend_elements = [Patch(facecolor='#2ecc71', edgecolor='black', label='Positive Effect'),
                   Patch(facecolor='#e74c3c', edgecolor='black', label='Negative Effect')]
plt.legend(handles=legend_elements, loc='lower right')

plt.tight_layout()
plt.show()

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FACTORIAL ANALYSIS: MAIN & INTERACTION EFFECTS (PARETO CHART)

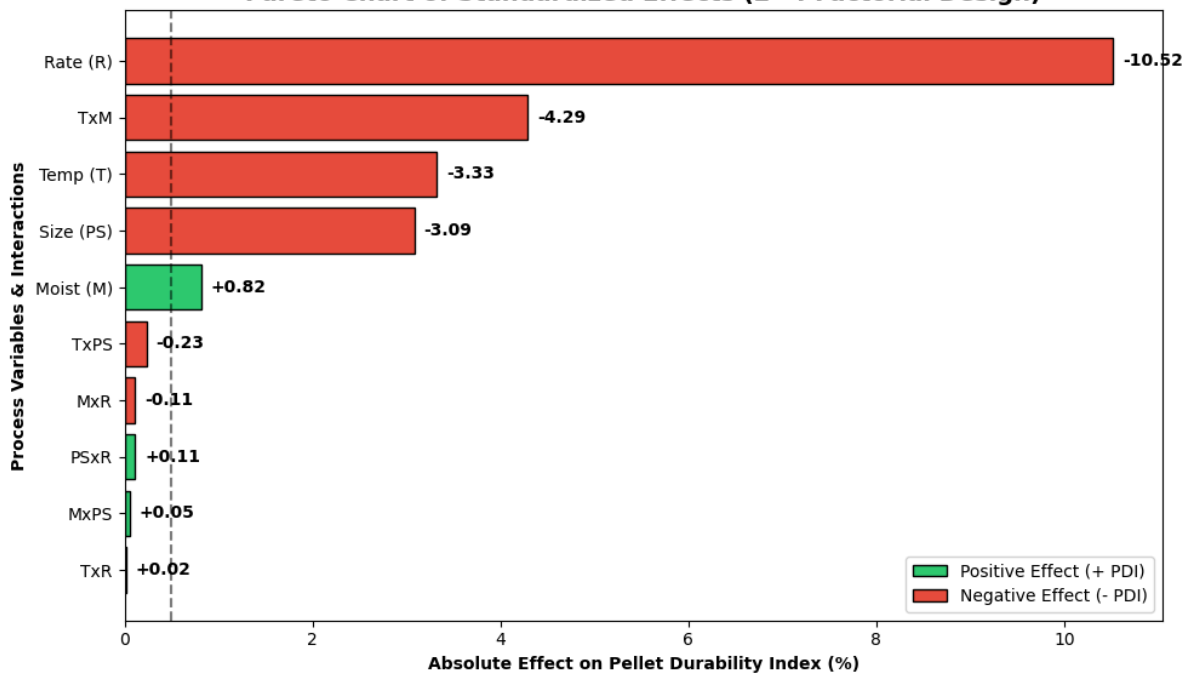
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SIMULATED 16-RUN FACTORIAL MATRIX (RAW FACTORY DATA)

Temp (T)	Moist (M)	Size (PS)	Rate (R)	Achieved_PDI
-1	-1	-1	-1	95.91
-1	-1	-1	1	85.22
-1	-1	1	-1	92.75
-1	-1	1	1	82.52
-1	1	-1	-1	100.85
-1	1	-1	1	90.35
-1	1	1	-1	98.19
-1	1	1	1	87.45
1	-1	-1	-1	96.84
1	-1	-1	1	86.64
1	-1	1	-1	93.64
1	-1	1	1	83.14
1	1	-1	-1	93.81
1	1	-1	1	82.67
1	1	1	-1	90.02
1	1	1	1	79.87

Calculated Factorial Effects (Ranked by Magnitude):

Rate (R) : 10.52% PDI swing (Decreases PDI)
TxM : 4.29% PDI swing (Decreases PDI)
Temp (T) : 3.33% PDI swing (Decreases PDI)
Size (PS) : 3.09% PDI swing (Decreases PDI)
Moist (M) : 0.82% PDI swing (Increases PDI)
TxPS : 0.23% PDI swing (Decreases PDI)
MxR : 0.11% PDI swing (Decreases PDI)
PSxR : 0.11% PDI swing (Increases PDI)
MxPS : 0.05% PDI swing (Increases PDI)
TxR : 0.02% PDI swing (Increases PDI)

Pareto Chart of Standardized Effects (2⁴ Factorial Design)

In []: